



A high-dimensional feature selection algorithm via fast dimensionality reduction and multi-objective differential evolution

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ABSTRACT

The multi-objective feature selection problem typically involves two key objectives: minimizing the number of selected features and maximizing classification performance. However, most multi-objective evolutionary algorithms (MOEAs) face challenges in high-dimensional datasets, including low search efficiency and potential loss of search space. To address these challenges, this paper proposes a hybrid algorithm based on fast dimensionality reduction and multi-objective differential evolution with redundant and preference processing (termed DR-RPMODE). In DR-RPMODE, the DR phase uses the freezing and activation operators to remove many irrelevant and redundant features in the high-dimensional datasets, thereby achieving fast dimensionality reduction. Subsequently, the RPMODE algorithm continues the search on the reduced datasets, improving the traditional differential evolutionary framework from two aspects: duplicated and redundant solutions are filtered by redundant handling, and a preference handling method that pays more attention to classification performance is designed for different preference objectives of decision-makers. In the experiment, DR-RPMODE is compared with seven feature selection algorithms on 16 classification datasets. The results indicate that DR-RPMODE outperforms the comparison algorithms on most datasets, demonstrating that it not only achieves outstanding optimization performance but also obtains good classification and scalability results.

1. Introduction

In real-world problems like machine learning and data mining, data often includes numerous features from diverse sources, such as sensors, surveys, social media, medical records, or transactions. While some features contribute significantly to model performance, others may be redundant or irrelevant, leading to challenges like overfitting and high computational complexity [1]. For instance, including irrelevant features, such as age when predicting a condition unrelated to age, can reduce model accuracy. To address these challenges, feature selection has become a crucial preprocessing step, aiming to only retain informative features from the original large feature set to improve model performance and generalization. [2].

Feature selection is a difficult task due mainly to a large search space, where the total number of possible solutions is $2^n - 1$ for a dataset with n features [3–6]. As the number of features increases, the search space grows exponentially, making the task more challenging. In most cases, searching for the optimal feature subset for a given dataset is practically impossible. Various search techniques have been

applied to feature selection, such as complete search, greedy search, heuristic search, and random search [7,8]. However, most existing feature selection methods still suffer from stagnation in local optima or high computational cost [9,10]. Unlike traditional random search methods, evolutionary computation (EC) can effectively explore the feature space by using the population and its evolution technology, which is very suitable for solving the feature selection problem.

In the classification problem, minimizing the number of selected features and maximizing the classification performance are key objectives in feature selection [11]. Minimizing features reduces complexity and prevents overfitting, while maximizing performance ensures the model retains sufficient information. Usually, they are conflicting and can be regarded as a multi-objective optimization problem. However, for datasets with high-dimensional features, many features may be redundant or irrelevant. Removing some features will not only not reduce classification performance but may even improve it [12,13]. Current multi-objective evolutionary algorithms (MOEAs) struggle with high-dimensional feature selection due to issues like premature convergence

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and poor solution diversity, particularly as the feature space grows exponentially, leading to many local optima [13]. Additionally, the computational cost of evaluating solutions increases with the number of features, making the problem harder to solve efficiently. To address these challenges, we propose a two-step approach: first, using a fast dimensionality reduction method to reduce the feature space, followed by MOEA-based feature selection. This reduces redundancy, decreases search space and computational cost, and allows the MOEA to focus on more relevant features, improving efficiency and solution quality.

After dimensionality reduction, feature selection can be viewed as a multi-objective optimization problem. Two key issues must be considered: the first is the differing preference between the two objectives. From the perspective of most decision-makers, maximizing classification performance is more important than minimizing the size of the feature subset [14]. To address this, we introduce constraints to balance the trade-off between feature subset size and classification performance. Specifically, we prioritize maximizing classification performance, the constraint ensures that only solutions achieving a certain level of Macro F1 score are considered valid. The second issue is redundancy. Since feature selection is a discrete optimization problem, solutions are represented by binary data (0s and 1s), which can lead to duplicated solutions. Redundant features may cause some solutions to select additional features that do not improve classification performance, resulting in redundant solutions. The presence of many duplicated and redundant solutions reduces population diversity, leading to premature convergence and suboptimal solutions. Addressing both types of redundancy is crucial for improving solution set diversity and algorithm's performance [14].

Differential evolution (DE) is a classical and popular method among EC methods. As pointed out in [15], many studies have shown that DE outperforms other evolutionary methods in many cases. DE has many advantages Compared to other methods that use evolutionary techniques to solve optimization problems. One is that it can be realized by adjusting only a few parameters, so it has been widely used in various fields. In addition, DE also stands out for its low space complexity.

This paper aims to address the challenges posed by high-dimensional datasets, which are prevalent in fields such as bioinformatics, image processing, and text mining. These datasets are often characterized by high feature dimensionality and uneven data distribution, leading to issues such as data redundancy, sparsity, increased computational complexity, and difficulties in convergence and analysis. To overcome these challenges, this paper proposes an efficient feature selection algorithm that balances solution convergence and diversity while ensuring good classification performance, scalability, and an improved ability to handle imbalanced datasets. The proposed hybrid algorithm, DR-RPMODE, integrates dimensionality reduction with multi-objective evolutionary optimization. Specifically, DR refers to the proposed fast dimensionality reduction algorithm, and RPMODE refers to the proposed multi-objective differential evolutionary algorithm. The main contributions of this paper can be summarized as follows:

1. A new algorithm combining dimensionality reduction and multi-objective evolution is introduced to effectively solve the problem of high-dimensional feature selection.

2. In DR, two novel operators are developed for fast dimensionality reduction: the freezing operator, which identifies irrelevant features, and the activation operator, which refines the feature subset by re-selecting features.

3. In RPMODE, a redundancy handling method is proposed. This method aims to remove duplicate and redundant solutions in the decision space, thereby promoting the diversity of the population.

4. In RPMODE, a preference handling approach is proposed to pay more attention to classification performance. This approach adds a constraint on the classification performance for each solution to treat solutions with different classification performances differently.

The rest of this paper is organized as follows. Section 2 introduces the basic work and existing EC-based high-dimensional feature

selection methods. Section 3 elaborates on the proposed algorithm DR-RPMODE. Section 4 shows the experimental setups. Section 5 presents the experimental results and analysis. Section 6 provides some further analysis. Section 7 concludes the paper and discusses future work.

2. Related work

2.1. Multi-objective optimization and multi-objective feature selection

Many problems usually need to optimize multiple objectives simultaneously, and these objectives are often contradictory in the optimization process. This kind of problem is called multi-objective optimization problem [16,17]. The minimization problem of multi-objective optimization can be mathematically defined as follows :

$$\begin{aligned} &\text{Minimize } F(x) = [f_1(x), f_2(x), \dots, f_m(x)] \\ &\text{Subject to: } \begin{cases} g_j(x) \leq 0, & j = 1, 2, \dots, p \\ h_j(x) = 0, & j = 1, 2, \dots, q \end{cases} \end{aligned} \quad (1)$$

where $x = [x_1, x_2, \dots, x_D]$ is a solution vector, m is the number of objectives, $g_j(x) \leq 0$ is the j th inequality constraint, $h_j(x) = 0$ is the j th equality constraint, and $[L_i, U_i]$ is the upper and lower bounds of the variable x_i .

The degree to which an individual x violates the j th constraint can be defined as:

$$\begin{cases} G_j(x) = \max \{0, g_j(x)\}, & 1 \leq j \leq p \\ H_j(x) = \max \{0, |h_j(x)| - \delta\}, & 1 \leq j \leq q \end{cases} \quad (2)$$

where δ is the tolerance for the equality constraint. Then, the degree to which x violates all constraints can be defined as:

$$CV(x) = \sum_{j=1}^p G_j(x) + \sum_{j=1}^q H_j(x) \quad (3)$$

A multi-objective feature selection problem can be defined as follows:

$$\begin{aligned} &\min F(x) = (f_{\text{ratio}}(x), f_{\text{loss}}(x))^T, \quad x = (x_1, x_2, \dots, x_D)^T \in \{0, 1\} \\ &\begin{cases} f_{\text{ratio}}(x) = \frac{\sum_{i=1}^D x_i}{D} \\ f_{\text{loss}}(x) = 1 - \frac{1}{c} \sum_{i=1}^c F1_i \end{cases} \end{aligned} \quad (4)$$

where D is the total number of features in a dataset, and x is the solution that represents a feature subset. $x_i = 1$ and $x_i = 0$ denote that the i th feature is selected and not selected. The first objective $f_{\text{ratio}}(x)$ is the selected feature ratio, which equals the ratio of the number of selected features to the total number of features. The second objective $f_{\text{loss}}(x)$ represents the Macro F1 loss, which is defined as 1 minus the Macro F1 score. The Macro F1 score is used to evaluate classification performance, where c is the number of classes of a dataset. $F1_i$ is the F1 score for the i th class.

2.2. Motivation

For feature selection problems, which are often discrete or combinatorial optimization problems, redundant or irrelevant features are a common and challenging issue, especially in high-dimensional feature selection. These redundant or irrelevant features increase computational complexity and lead to model overfitting, thereby reducing the model's generalization ability and performance. However, identifying and removing these features is not an easy task. Therefore, both the DR and the redundancy handling in RPMODE proposed are designed to address this issue.

Moreover, further improvements in balancing conflicting goals, such as maximizing classification performance while minimizing the number of selected features, are needed in the current multi-objective

feature selection methods [4]. Typically, maximizing classification performance is considered more important than minimizing the number of selected features, as classification performance directly impacts the practical application of the model. To distinguish between solutions with different classification performances, we propose a preference handling method, applying greater pressure to solutions with poorer classification performance, making them less likely to survive in the population, and thereby more effectively optimizing classification performance.

2.3. Existing EC-based high-dimensional feature selection methods

In recent years, researchers have utilized EC methods for feature selection in high-dimensional datasets. So far, many evolutionary algorithms have been applied to feature selection problems [18], including genetic algorithms (GAs) [19], particle swarm optimization (PSO) [20], DE [21], and others. This section briefly reviews some previously proposed EC-based methods for high-dimensional feature selection.

2.3.1. Feature selection methods based on single-objective evolution

Many researchers use classification error rate as the fitness function to evaluate the quality of selected feature subsets. For example, a multi-label feature selection algorithm based on the Competitive Swarm Optimizer is proposed in [22]. This algorithm selects the most salient features based on their redundancy and relevance by using the MI Criterion to construct half of the initial population and applying an objective function to minimize loss. In [23], a novel method integrates Ant Colony Optimization (ACO) with Temporal Difference (TD) reinforcement learning to optimize multi-label feature selection by learning heuristic functions from experiences. The SemiACO algorithm computes feature relevancy concerning class labels and measures feature redundancy, combining nonlinear reinforcement learning and ACO heuristic functions to address both supervised and unsupervised learning tasks in feature selection [24]. Another approach, SFE-PSO, is proposed in [13], which combines dimensionality reduction and optimization. SFE first removes irrelevant features to reduce the feature space, and then PSO optimizes the reduced set. Experimental results show that SFE-PSO is effective for high-dimensional data. However, treating feature selection as a single-objective optimization problem has its limitations. It often focuses solely on optimizing classification performance, neglecting other important aspects such as solution diversity and feature interaction. This may result in selecting subsets of features that overfit the model or ignore the multi-dimensional nature of the problem.

2.3.2. Traditional feature selection methods based on multi-objective evolution

Given the superior performance of MOEAs, in recent years, more and more MOEA-based feature selection methods have been developed to solve multi-objective feature selection problems [25–27]. In [28], a multi-objective particle swarm optimization algorithm is proposed, which extends the multi-objective particle swarm optimization algorithm [29] to handle multi-objective feature selection problems. Experimental results demonstrate that this method outperforms single-objective feature selection methods and some typical MOEAs. In [30], a multi-objective differential evolution algorithm was proposed for feature selection. This method embeds probabilistic binary mutation and single-bit net cleansing search based on probability difference into the differential evolution algorithm to enhance its search capability. However, traditional MOEAs, due to their lack of specialized design for feature selection, are prone to challenges such as computational complexity and maintaining solution diversity when dealing with high-dimensional feature selection problems.

2.3.3. Advanced feature selection methods based on multi-objective evolution

Today, more and more researchers have designed MOEAs specifically to solve high-dimensional feature selection problems [31,32]. The Duplication Analysis Evolutionary Algorithm (DAEA) [33] tackles redundancy in multi-objective feature selection by removing similar solutions to enhance population diversity. It also limits the number of features selected during initialization. While it performs well on many high-dimensional datasets, its performance tends to deteriorate as the dimensionality increases. In [34], a problem reformulation and constraint handling method are designed to address diverse preferences among objectives. While it achieves good classification performance, it does not strike an optimal balance between convergence and diversity. In [35], the Multi-Filtration Feature Selection (MFFS) method is introduced, which includes feature extraction, subset selection, ranking, and classification. While it demonstrates good diversity in the solution set, its highest classification performance is relatively lower compared to other methods. In [36], Liao et al. introduced parallel mechanisms to MOEA/D-STAT for feature selection, comparing master-slave and island parallelization. Master-slave parallelization achieves better speedup and computational efficiency but tends to produce relatively larger feature subsets, while island parallelization enhances solution diversity but suffers from reduced performance and limited search capabilities. Khammassi et al. [37] proposed a MOEA using NSGA-II and LR for network intrusion detection, showing improved accuracy and feature reduction. However, their scalability is limited due to the algorithms being designed for specific dataset types. In [38], MOEA-NZD is designed for sparse large-scale multi-objective optimization problems, using all-zero initialization and clustering to promote sparsity. It adopts a two-stage process: the first stage accelerates convergence to identify key features, while the second enhances non-zero variable mutations to refine feature subsets. This approach is well-suited for high-dimensional feature selection. We have included a comparative analysis table (see Table 1) that summarizes the advantages and disadvantages of the reviewed advanced MOEAs.

3. Proposed method

This section is divided into two parts to illustrate the proposed hybrid algorithm DR-RPMODE. Section 3.1 elaborates on DR, while Section 3.2 elaborates on RPMODE.

3.1. DR

Similar to the EC methods, the proposed method DR also searches the decision space by iteratively searching for solutions. However, unlike population-based evolutionary algorithms, DR only uses a search solution, which can significantly reduce computational costs and memory consumption. Feature selection in this method is a simple binary problem: each feature can either be selected or not. In our algorithm, the solution X represents the feature selection result, with each feature's state (selected or not) indicated by a binary value. This solution is encoded as a binary vector, as shown in Fig. 1, where $x_i = 0$ indicates that the i th feature is in the non-selected state, and $x_i = 1$ indicates that the feature is in the selected state.

The algorithm uses two main operations: freezing and activation. The freezing operator removes features by changing them from selected to non-selected, while the activation operator adds features by changing them from non-selected to selected. These operations balance the exploration and exploitation of feature subsets to improve the solution X during the search process. To assess the quality of a solution, we use Macro F1 score as the fitness function. This is defined as follows:

$$fit = \frac{1}{c} \sum_{i=1}^c F1_i \quad (5)$$

where c is the number of classes in the dataset, and $F1_i$ is the F1 score for the i th class.

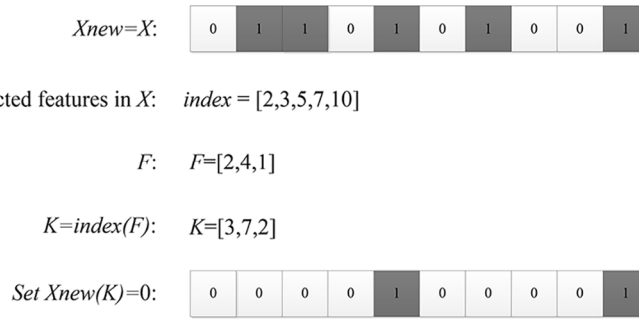


Fig. 2. Application of the freezing operator.

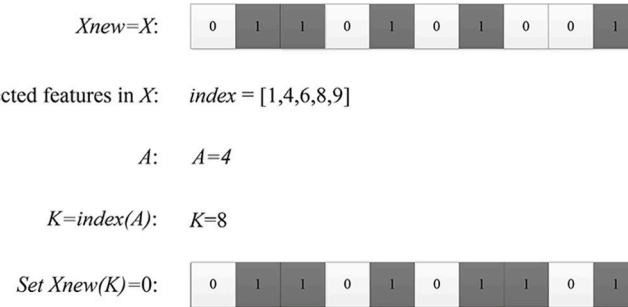


Fig. 3. Application of the activation operator.

in Algorithm 1). The activation operator helps the algorithm re-select previously unselected features. This process is called “activation”, and the number of features to be re-selected is AN . Since the algorithm may select only a few features after multiple search steps, using larger values for AN may not achieve the desired improvement effect [13]. Therefore, this paper sets AN to 1.

As shown in Fig. 3, the first case is handled; that is, X is not updated for a long time. Suppose $AN = 1$, the generated pseudo-random number A is 4, and then access the index vector of features in the non-selected state, obtain 8. Therefore, the feature indexed as 8 in X will be activated.

3.2. RPMODE

The RPMODE algorithm continues to search in the new search space S , as shown in Algorithm 2. It begins by creating a population Pop with N solutions, where the number of selected features in each solution is randomly initialized (line 1). Each generation in the main loop of RPMODE consists of three operations: mating selection, offspring reproduction, and environmental selection. Mating selection aims to choose excellent parent individuals from the population to generate offspring individuals, employing a tournament selection method based on non-dominated sorting and crowding distance of the Pareto front (line 4); offspring reproduction utilizes the DE algorithm with polynomial mutation (line 5, detailed in Section 3.2.1); finally, the environmental selection (lines 6 to 9) selects elite solutions to survive from the union of parent and offspring populations, mainly involving redundancy handling and preference handling (to be detailed in Sections 3.2.2 and 3.2.3, respectively). After the loop ends, the nondominated solution set is output from the final population (line 11).

3.2.1. Offspring reproduction

Through mating selection, the mating pool produces 2^*N excellent individuals. The first N individuals are assigned to population $Parent_2$, while the last N individuals are assigned to population $Parent_3$. The parent population is $Parent_1$.

Algorithm 2 RPMODE

Require: population size N ; decision space dimensionality D ; termination criterion τ ;

Ensure: non-dominated solutions PF ;

- 1: $Pop \leftarrow \text{InitializePopulation}()$ ▷ Population initialization
- 2: $Pop \leftarrow \text{Ndsort}(Pop, N)$ ▷ Non-dominated sorting
- 3: **while** $t < \tau$ **do**
- 4: $MatingPool \leftarrow \text{TournamentSelection}(Pop)$
- 5: $Off \leftarrow \text{OperatorDE}(Pop, MatingPool)$
- 6: $Par_Off1 \leftarrow Pop \cup Off$ ▷ Union of population and offspring
- 7: $Par_Off2 \leftarrow \text{RedundantHandling}(Par_Off1)$
- 8: $Par_Off3 \leftarrow \text{PreferenceHandling}(Par_Off2)$
- 9: $Pop \leftarrow \text{Ndsort}(Par_Off3, N)$ ▷ Non-dominated sorting
- 10: **end while**
- 11: $PF \leftarrow \text{Select non-dominated solutions in } Pop$

The process of generating the i th solution v^i by DE is as follows :

$$v^i(j) = \begin{cases} Parent_1^i(j) + F \times (Parent_2^i(j) - Parent_3^i(j)), & rand \leq CR \\ x^i(j), & \text{otherwise} \end{cases} \quad (8)$$

where i is the i th individual, j is the j th component, F is mutation factor, CR is crossover rate, and $rand$ is a random number uniformly distributed in the range $[0,1]$.

Then, the offspring solutions x_i are generated for v^i according to polynomial mutation (Eq. (9)).

$$x^i(j) = \begin{cases} v^i(j) + \sigma(j) \times (b(j) - a(j)), & rand \leq p \\ v^i(j), & \text{otherwise} \end{cases} \quad (9)$$

The parameter $\sigma(j)$ can be obtained according to Eq. (10):

$$\sigma(j) = \begin{cases} (2 \times rand)^{\frac{1}{\eta+1}} - 1, & rand < 0.5 \\ 1 - (2 - 2 \times rand)^{\frac{1}{\eta+1}}, & \text{otherwise} \end{cases} \quad (10)$$

Before repair:

0.3	1.1	0.7	0.4	0.8	-0.2	1.2	0.1	0.5	1
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After repair:

0	1	1	0	1	0	1	0	0	1
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Fig. 4. Individual repair strategy.

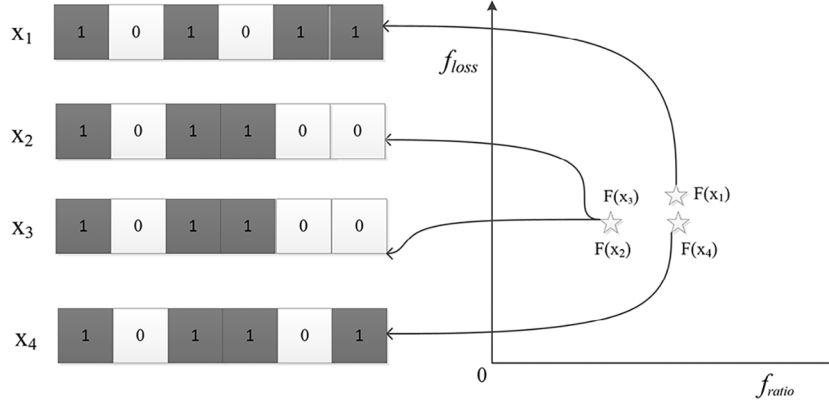


Fig. 5. Redundancy analysis.

where η is the distribution index, p is the mutation probability. $a(j)$ and $b(j)$ are the lower and upper bounds of the j th decision variable, respectively.

After DE and polynomial mutation, the individual solutions are transformed into continuous real-valued encodings. To convert these continuous values into a binary decision space, a threshold Δ is used to compare with each component. A feature is selected only if its corresponding value is greater than the threshold Δ . To limit the number of selected features and ensure the sparsity of the subset, Δ should be set to greater than 0.5 [1]. This paper sets Δ to 0.6. Fig. 4 provides an example of the individual repair strategy.

3.2.2. Redundancy handling

Redundancies are classified into two types: the first type is duplicated solutions, which are solutions that select the same features. The second type is redundant solutions, which are solutions that select additional features compared to non-dominated solutions but do not improve classification performance. Compared to duplicated solutions, which can be easily identified and removed, redundant solutions are more challenging to handle.

Fig. 5 provides an example to facilitate understanding of the proposed redundancy handling method. Suppose that solutions x_2 and x_3 are two duplicate non-dominated solutions in the search space, while x_1 and x_4 are dominated solutions. The first step is to remove the duplicated solution x_3 and retain x_2 . Compared to x_2 , both x_1 and x_4 select more features but do not reduce the Macro F1 Loss. To determine which solution is redundant, we need to calculate the Hamming distance between x_2 and x_1 , x_4 at the positions where x_2 is 1. The Hamming distance measures the similarity between two solutions at specified positions. $d_{ham}(x_4, x_2) = 0$ indicates that x_4 selects additional features, so x_4 is a redundant solution and should be deleted, while $d_{ham}(x_1, x_2) = 1$ indicates x_1 does not select additional features, so x_1 is not a redundant solution and should be retained.

3.2.3. Preference handling

After redundancy processing, we analyze the distribution of the remaining solutions. Fig. 6 provides an example of how solutions are

distributed in the objective space. In Fig. 6(a), we define x_t as the critical point, representing the solution with the maximum Macro F1 Loss among all non-dominated solutions, i.e., $x_t = \arg \max_{x \in F_1} f_{loss}(x)$, where F_1 is the first Pareto front after non-dominated sorting [39]. Among all non-dominated solutions, x_t selects the fewest features but exhibits the worst classification performance. Given the focus on improving classification performance, the Macro F1 Loss of x_t is considered as a threshold. Solutions with a Macro F1 Loss higher than this threshold are deemed infeasible. As a result, the solutions are divided into two categories: the red points and the green points in Fig. 6(b). The red points, which have higher Macro F1 Loss than x_t , are considered infeasible solutions, while the green points, with Macro F1 Loss below the threshold, are deemed feasible solutions.

By separating the solution space into feasible and infeasible solutions, the multi-objective optimization problem can be reformulated as a constrained multi-objective problem. The redefined feature selection problem can be written as:

$$\begin{aligned} \min F(x) &= (f_{ratio}(x), f_{loss}(x))^T \\ \begin{cases} f_{ratio}(x) = \frac{\sum_{i=1}^D x_i}{D} \\ f_{loss}(x) = 1 - \frac{1}{c} \sum_{i=1}^c F1_i \\ g(x) = f_{loss}(x) - f_{loss}(x_t) \leq 0, \quad x_t = \arg \max_{x \in F_1} f_{loss}(x) \end{cases} \end{aligned} \quad (11)$$

where x_t is the critical point, which represents the solution with the highest Macro F1 Loss among all non-dominated solutions. The inequality constraint $g(x)$ signifies that if the Macro F1 Loss of solution x is less than the Macro F1 Loss of x_t , then the solution is feasible ($g(x) \leq 0$); otherwise, it is infeasible ($g(x) > 0$). It is worth noting that the position of the critical point x_t may change with the distribution of each generation's population in the objective space, as the corresponding non-dominated solutions also change. Therefore, this can also be viewed as a dynamic constrained multi-objective problem.

To address this dynamic constraint problem, we use the penalty function method [40], a well-known technique for handling constraints. The idea is to add a penalty term to the objective function, which increases as a solution violates the constraint. This penalty term combines

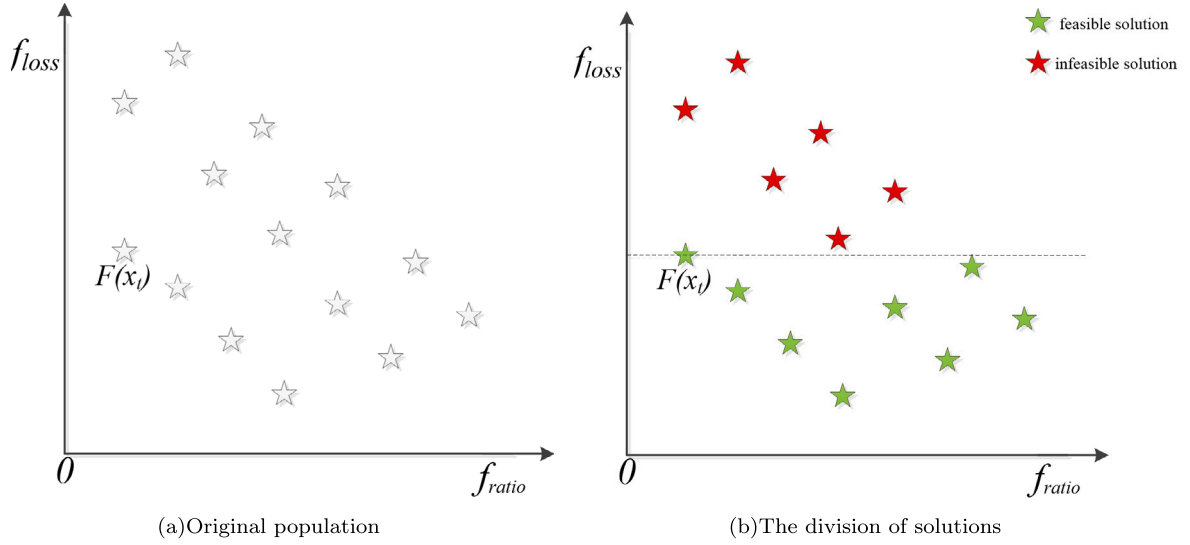


Fig. 6. Distribution of the population.

the original objective function using a penalty factor r . In the early stages of the algorithm, the penalty is small, allowing some infeasible solutions that could lead to better results. This helps maintain diversity in the population. As the algorithm progresses, the penalty increases, focusing more on finding feasible solutions. An adaptive strategy is used to control r throughout the process. The reconstructed objective function is represented as follows:

$$\begin{cases} f_{ratio}(x) = \frac{\sum_{i=1}^D x_i}{D} + r \times CV(x) \\ f_{loss}(x) = 1 - \frac{1}{c} \sum_{i=1}^c F1_i + r \times CV(x) \end{cases} \quad (12)$$

$$r = \frac{g}{G}, \quad CV(x) = \max\{0, g(x)\}$$

where $CV(X)$ is the degree of constraint violation of the solution x , g is the current iteration number of the population, and G is the maximum iteration number of the population.

3.3. Parameters analysis and roles

This section discusses the role of key parameters in the proposed algorithm and how they control different aspects of the search process, including exploration, exploitation, convergence, diversity, and encoding effects.

In the proposed DR algorithm, the parameters play critical roles in balancing exploration, exploitation, convergence, and diversity. Exploration is driven by FR , which controls the number of features to remove, and FES , which influences the search duration, allowing more exploration in the early stages. The activation operator focuses on exploitation by reintroducing important features to refine the solutions. Convergence is guided by the fitness function and FES , ensuring the algorithm moves towards an optimal solution. To maintain diversity and avoid premature convergence, the FR and early termination conditions are employed. The binary encoding simplifies the problem, providing computational efficiency, though it may lack the fine-grained control of other encodings. Lastly, the freezing and activation operators work together to enhance the search power, balancing exploration and exploitation to find the optimal feature subset effectively.

In the RPMODE algorithm, exploration is driven by the DE, where parameters like F and CR encourage the generation of diverse solutions. Exploitation is achieved through environmental selection, which retains the most promising solutions based on fitness. The non-dominated sorting mechanism ensures convergence by gradually focusing on optimal solutions while maintaining diversity through crowding distance and redundancy handling. Overall, the search operators in

RPMODE, such as mutation, crossover, and environmental selection, enhance the algorithm's ability to explore the search space, refine solutions, and ultimately converge to an optimal feature subset while maintaining diversity.

4. Experiment setting

4.1. Datasets and classifiers

The experimental process utilized 16 classification datasets from the UCI Machine Learning Repository [41] and the scikit-feature feature selection repository [42], as detailed in Table 2. The number of features ranged from 166 to 24482, the number of classes varied from 2 to 40, and the number of instances ranged from 50 to 2000. The original datasets are split into 70% training and 30% testing sets.

For classifier selection, KNN is often chosen by many researchers because of its easy implementation and intuitive understanding [43]. However, it has a notable drawback: it may lead to classification errors when dealing with imbalanced datasets. Therefore, this study adopts a weighted KNN method proposed by Hechenbichler et al. [44], which assigns different weights to K nearest neighbor samples based on the distances between each neighbor sample and the test sample. The closer the distance, the larger the weight assigned; conversely, the farther the distance, the smaller the weight assigned. This weighted K -nearest neighbor method is less sensitive to the choice of K value, demonstrating good robustness. To ensure that the selected K value adapts to the characteristics of different datasets, we used 10-fold cross-validation for hyperparameter tuning. Specifically, the dataset is first divided into 10 subsets, with 9 subsets used for training and 1 subset used for validation. In each fold, we train the model and evaluate its Macro F1 score as the performance metric. We tested multiple candidate K values ($K = 1, 3, 5, 7, 9$) and computed the average Macro F1 score for each K value across all cross-validation folds. Ultimately, we selected the K value with the best average Macro F1 score as the model's hyperparameter.

4.2. Comparison algorithm

In this paper, two traditional MOEAs, four advanced MOEAs, and RPMODE without DR processing are compared with DR-RPMODE. The two traditional MOEAs are NSGA-II [39] and MOEA/D [45]. The four advanced MOEAs are MOEA-NZD [38], DAEA [33], MFFS [46], and PRDH [34]. NSGA-II is the most classical dominance-based MOEA and

Table 2

Datasets sorted by the number of features.

No.	Dataset	Features	Instances	Classes
1	MUSK1	166	476	2
2	Madelon	500	2000	2
3	Isolet	617	1560	26
4	ORL	1024	400	40
5	Colon	2000	62	2
6	Lung	3312	203	4
7	Basehock	4862	1993	2
8	9tumor	5726	295	9
9	Tox_171	5748	171	4
10	Drivface	6400	606	3
11	Allaml	7129	72	2
12	Orlraws10p	10304	100	10
13	Brain_Tumor_2	10367	50	4
14	CLL_SUB_111	11340	111	3
15	Smk_can_187	19993	187	2
16	Breast_Cancer	24482	97	2

Table 3

Comparison algorithms and parameter settings.

Alg	Parameters
NSGA - II	$P_c = 0.9, P_m = 1/D$
MOEA/D	$P_c = 0.9, P_m = 1/D, \theta = 0.6$
MOEA-NZD	$P_c = 0.9, P_m = 1/D$
DAEA	$T = \max(40, 2N), P_{c1} = 0.8, P_{c2} = 0.8, P_m = 0.8$
MFFS	$P_c = 1, P_m = 1/D, \beta_c = 20, \beta_m = 20$
PRDH	$P_c = 0.9, P_m = 1/D$
RPMODE	$CR = 0.9, F = 0.5$
DR-RPMODE	$FR_{max} = 0.3, FR_{min} = 0.001, CR = 0.9, F = 0.5$

MOEA/D is the most classical decomposition-based MOEA. MOEA-NZD is designed for sparse large-scale multi-objective optimization problems, making it ideal for high-dimensional feature selection. DAEA, MFFS, and PRDH are MOEAs specifically designed for feature selection proposed in recent years. All comparative experiments are conducted on the open-source platform PlatEMO [47].

Traditional MOEAs like NSGA-II and MOEA/D in the PlatEMO platform use equal probability distribution for population initialization, leading to feature subsets with many features. However, feature selection typically aims for sparse subsets, where only a few features are selected. To improve experiment effectiveness, we applied RPMODE's initialization method to NSGA-II and MOEA/D. Specifically, the number of ones in each decision vector will be randomly distributed. Additionally, to avoid high computational costs on large sample datasets, we set an instance threshold to 1000. If the number of instances in a dataset exceeds this threshold, a random subset of 1000 instances is selected to replace the original dataset. Each MOEA runs independently 30 times, the maximum objective function evaluation times $maxFEs$ is 10000, and the population number N is 100. Each compared MOEA's customizable parameters follow their original papers, as detailed in Table 3.

4.3. Evaluating indicator

In this paper, five performance indicators are used to evaluate the performance of eight MOEAs. They are the Inverted Generational Distance (IGD) [48], Hypervolume (HV) [49], Macro F1 score, Average Selected Feature Number (AFN), and algorithm runtime.

IGD evaluates algorithms' convergence and distribution performance by computing the minimum distance from each solution on the true Pareto front to the set of non-dominated solutions obtained by the algorithm [48]. When calculating IGD, finding the true Pareto front is difficult because the true Pareto front of feature selection in each data set is usually unknown. Therefore, we collected the non-dominated solution sets from 30 runs of each algorithm on each dataset

and considered the non-dominated solutions in the union of these sets as the true Pareto front.

HV evaluates algorithms' convergence and distribution performance by computing the region's volume in the objective space enclosed by the non-dominated solution set obtained by the algorithm and a reference point [50]. For HV, firstly, all objective values obtained by each algorithm are normalized to the scale [0,1] by the ideal point and the lowest point in each target direction [33], and the ideal point and the lowest point are set to (0,0) and the maximum value in each direction of the real Pareto front. The reference point of HV is set to (1,1).

AFN refers to the average number of features in the feature subset that achieves the minimum Macro F1 Loss in multiple independent runs of the algorithm. This metric measures the average size of the selected feature subset when evaluating algorithm performance.

The Wilcoxon test with a confidence level of 95% is used to assess whether the performance of DR-RPMODE differs significantly from the comparison algorithms for the IGD and HV metrics [51]. Additionally, the Friedman test is conducted for HV, IGD, Macro F1 score, and AFN to compare the effectiveness of the DR-RPMODE algorithm with other algorithms.

5. Experimental results and analysis

5.1. Median pareto front analysis

To visually demonstrate the advantages of DR-RPMODE in terms of convergence and diversity, the median Pareto fronts of all comparison algorithms on 8 different dimensions of test sets are shown in Fig. 7. The median Pareto front refers to the distributions of nondominated feature subsets obtained by each algorithm on test sets in terms of median HV value, which can reflect the algorithm's general performance. The selected 8 datasets are MUSK1, Isolet, Colon, BASEHOCK, TOX_171, Allaml, Brain_Tumor_2, and Smk_can_187. The selection rule is as follows: from the 16 datasets, those at odd indices (i.e., the 1st, 3rd, 5th, 7th, 9th, 11th, 13th, and 15th datasets) were chosen for plotting to ensure the dataset distribution's representativeness.

Fig. 7 shows that among these eight MOEAs, the non-dominated solution sets obtained by DR-RPMODE are competitive in terms of both classification performance and solution distributions, preserving more Pareto-optimal solutions and demonstrating better convergence. This performance is primarily attributed to the following factors: The fast dimensionality reduction strategy DR, which enhances convergence by efficiently eliminating irrelevant features; The preference handling technique improves classification performance, while the redundancy handling technique increases solution diversity by addressing duplicate solutions. Compared to RPMODE, the solution sets obtained by DR-RPMODE are more convergent, demonstrating that the DR strategy effectively removes a large number of redundant solutions while preserving important features. Furthermore, applying RPMODE's population initialization method to NSGA-II, MOEA/D, and MOEA-NZD results in a non-dominated solution set that does not cluster in a small region with numerous features, as described in [34]. Instead, it demonstrates good classification performance and distribution of solution sets.

As shown in Fig. 7, DR-RPMODE significantly reduces Macro F1 Loss as the feature subset size increases across all datasets, indicating that increasing the number of features is both effective and worthwhile. Although in some datasets (such as Isolet), the rate of decrease in Macro F1 Loss diminishes as the feature subset size increases, the reduction is still substantial, suggesting that the added features remain effective. Once Macro F1 Loss reaches its minimum (which reaches 0 in many datasets), further increasing the feature subset size will not reduce Macro F1 Loss, indicating that the features added are not informative and discriminative enough, from a cost-benefit perspective, continuing to increase the feature subset size is no longer worthwhile. In most

Table 4

Comparative results of HV values (mean and standard deviation) obtained by eight algorithms on the test sets.

Problem	NSGAII	MOEA/D	MOEA-NZD	DAEA	MFFS	PRDH	RPMODE	DR-RPMODE
MUSK1	1.7355e-1(0.00e+0)=	5.4545e-2(4.98e-2)+	1.7355e-1(0.00e+0)=	1.7280e-1(4.14e-3)=	1.7355e-1(0.00e+0)=	9.0909e-2(7.06e-7)+	1.7355e-1(0.00e+0)=	1.7355e-1(0.00e+0)=
Madelon	6.9007e-1(5.26e-3)=	3.7049e-2(8.28e-2)+	6.6537e-1(1.86e-1)=	6.9326e-1(4.51e-3)=	6.5662e-1(3.79e-3)=	6.7455e-1(4.91e-3)=	6.2660e-1(2.22e-2)+	6.7257e-1(3.45e-2)
Isotlet	8.4495e-1(5.45e-3)=	6.5359e-1(3.55e-2)+	8.5021e-1(2.63e-2)=	8.6682e-1(3.33e-3)=	8.2377e-1(2.40e-3)+	8.3940e-1(5.31e-3)+	8.3217e-1(2.46e-2)+	8.5548e-1(2.57e-2)
Orl	7.1783e-1(2.12e-2)=	0.000e+0(0.00e+0)+	7.2284e-1(2.66e-1)=	7.2577e-1(4.91e-3)=	6.9832e-1(4.05e-2)=	7.0030e-1(2.19e-2)+	7.1474e-1(2.44e-2)=	7.3128e-1(2.32e-2)
Colon	4.5242e-1(2.47e-2)+	1.3692e-1(3.32e-2)+	5.3428e-1(4.12e-1)=	5.3581e-1(9.21e-3)+	5.5234e-1(1.94e-2)=	4.6592e-1(2.17e-2)+	4.1415e-1(1.88e-1)+	5.7305e-1(1.73e-2)
Lung	7.3105e-1(1.58e-2)=	0.000e+0(0.00e+0)+	7.6307e-1(2.55e-1)=	7.4592e-1(1.04e-2)=	7.4011e-1(1.02e-2)=	7.2218e-1(1.03e-2)=	7.2644e-1(2.21e-2)=	7.4377e-1(1.36e-2)
BASEHOCK	8.9723e-1(2.33e-2)+	0.000e+0(0.00e+0)+	9.3364e-1(4.12e-3)=	9.4478e-1(7.36e-3)=	9.2355e-1(2.25e-2)=	9.3212e-1(4.39e-3)=	9.3105e-1(1.52e-2)=	9.4933e-1(1.26e-1)
9Tumor	5.9311e-1(5.33e-2)=	0.000e+0(0.00e+0)+	5.9733e-1(1.77e-1)+	6.2522e-1(2.55e-2)=	5.4319e-1(3.26e-2)+	5.7554e-1(3.66e-2)=	5.8210e-1(2.39e-2)=	6.2177e-1(1.42e-2)
TOX_171	6.9555e-1(2.33e-2)=	8.6303e-2(1.75e-2)+	6.1892e-1(2.37e-1)=	6.7930e-1(2.70e-2)=	6.1786e-1(3.24e-2)+	6.1279e-1(4.82e-2)+	6.3665e-1(4.69e-2)+	6.7490e-1(2.72e-2)
DrivFace	7.9263e-1(1.63e-3)=	0.000e+0(0.00e+0)+	7.1207e-1(2.37e-2)+	7.9672e-1(1.55e-3)=	8.0264e-1(2.29e-3)=	7.8952e-1(1.62e-3)=	7.9212e-1(3.47e-2)=	7.9563e-1(2.68e-1)
Allaml	1.5563e-1(3.63e-2)+	0.000e+0(0.00e+0)+	1.9028e-1(3.16e-2)+	1.6602e-1(2.48e-3)+	1.3302e-1(0.00e+0)+	1.6629e-1(3.41e-3)+	1.5103e-1(4.73e-3)+	7.9165e-1(3.14e-2)
Orlraws10p	4.7134e-1(3.25e-2)=	0.000e+0(0.00e+0)+	4.5507e-1(5.33e-2)=	4.5773e-1(6.28e-2)=	4.2894e-1(5.44e-2)=	4.3429e-1(3.68e-2)=	4.2931e-1(4.55e-2)=	4.2352e-1(9.32e-3)
Brain_Tumor_2	3.6684e-1(4.57e-2)+	0.000e+0(0.00e+0)+	3.2949e-1(7.40e-2)+	3.9435e-1(3.12e-2)+	3.7129e-1(1.46e-2)+	3.1355e-1(3.37e-2)+	3.5638e-1(4.22e-2)+	8.7782e-1(6.36e-3)
CLL_SUB_111	3.6755e-1(5.25e-2)+	0.000e+0(0.00e+0)+	3.9520e-2(6.42e-2)+	4.7852e-1(4.11e-2)+	3.6475e-1(9.08e-3)+	2.1175e-1(3.41e-1)+	3.3858e-1(2.18e-2)+	7.8353e-1(2.48e-2)
Smk_can_187	8.2057e-1(2.27e-2)+	0.000e+0(0.00e+0)+	7.4265e-1(6.46e-2)+	8.4753e-1(4.00e-2)+	8.1544e-1(3.12e-2)+	8.4744e-1(5.45e-2)=	8.4826e-1(2.53e-2)=	8.5013e-1(7.17e-2)
Breast_Cancer	6.7443e-1(6.54e-2)+	0.000e+0(0.00e+0)+	6.2165e-1(5.43e-1)+	6.7354e-1(2.21e-2)+	4.5253e-1(2.25e-1)+	6.7305e-1(3.58e-2)+	6.9422e-1(4.93e-2)+	9.2218e-1(1.73e-2)
+/-/=	7/0/9	16/0/0	7/0/9	6/0/10	8/0/8	9/0/7	8/0/8	-
Rank1	3.4071	7.5516	3.8438	2.7507	4.1128	4.4840	4.2754	2.1135

Table 5

Comparative results of IGD values (mean and standard deviation) obtained by eight algorithms on the test sets.

Problem	NSGAII	MOEA/D	MOEA-NZD	DAEA	MFFS	PRDH	RPMODE	DR-RPMODE
MUSK1	0.0000e+0(0.00e+0)=	8.0320e-2(2.12e-4)+	0.0000e+0(0.00e+0)=	4.7295e-2(7.32e-4)=	4.8585e-2(2.68e-3)=	7.5695e-2(2.12e-3)+	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=
Madelon	2.5478e-2(4.58e-3)+	1.6755e-1(2.14e-2)+	2.2475e-2(7.71e-2)=	1.5247e-2(3.25e-3)=	1.5417e-2(2.40e-3)=	3.3401e-2(3.17e-3)+	1.6750e-2(2.44e-3)=	1.5612e-2(5.20e-2)
Isotlet	4.2517e-2(2.47e-2)+	2.6312e-1(9.25e-3)+	6.2721e-2(2.66e-2)+	8.4758e-3(4.41e-3)=	8.4758e-3(4.41e-3)=	6.4018e-2(2.65e-2)+	4.1277e-2(3.41e-2)+	8.9055e-3(4.15e-3)
Orl	1.8692e-2(4.05e-3)=	2.4204e-1(3.04e-3)+	1.2359e-2(7.92e-2)=	1.7838e-2(1.67e-3)=	1.7412e-2(5.27e-3)=	3.4672e-2(9.05e-4)+	1.6291e-2(7.13e-2)=	1.6881e-2(1.60e-2)
Colon	5.9473e-2(1.60e-2)+	5.7982e-1(2.96e-2)+	6.0822e-1(9.16e-3)+	5.5487e-2(2.08e-3)+	9.8207e-2(1.72e-4)+	7.7067e-2(3.55e-3)+	4.8331e-2(2.04e-2)=	4.6113e-2(2.16e-3)
Lung	9.8972e-3(3.19e-3)=	1.8527e-1(2.52e-3)+	9.5968e-3(2.19e-2)=	1.2304e-2(3.28e-3)+	8.6689e-3(3.43e-3)=	1.3265e-2(3.18e-3)+	1.1295e-2(4.57e-3)+	9.9345e-3(3.62e-3)
BASEHOCK	4.1673e-2(1.20e-2)+	8.1524e-2(1.72e-3)+	3.7532e-2(3.01e-2)+	1.7749e-2(2.12e-3)=	2.7143e-2(1.13e-2)+	5.1538e-2(6.72e-3)+	1.7683e-2(4.32e-3)=	1.6571e-2(5.91e-3)
9Tumor	5.3553e-2(2.94e-2)+	2.2943e-1(3.01e-3)+	5.5344e-2(5.77e-2)+	1.8146e-2(1.38e-4)=	4.1442e-2(3.92e-3)+	3.3672e-2(2.92e-2)=	3.3221e-2(1.87e-2)=	2.9603e-2(1.06e-2)
TOX_171	4.5839e-2(1.39e-2)+	2.6524e-1(1.02e-2)+	7.2295e-2(3.97e-2)=	1.3631e-2(5.68e-3)=	7.1065e-3(1.95e-3)+	8.9202e-2(2.65e-2)+	2.0302e-2(5.61e-3)=	1.4373e-2(3.97e-3)
DrivFace	3.4175e-3(2.02e-3)=	5.7412e-2(4.64e-3)+	3.6244e-3(6.36e-2)=	4.1673e-3(2.40e-3)=	3.9374e-3(2.18e-4)=	1.7720e-2(7.17e-3)+	5.9442e-3(1.81e-3)+	3.0217e-3(5.11e-3)
Allaml	1.8268e-2(3.45e-1)+	8.1934e-1(2.78e-2)+	5.1124e-2(4.20e-2)+	1.2175e-2(2.13e-3)=	3.1788e-2(3.42e-3)+	2.1614e-2(1.74e-2)=	1.9543e-2(2.63e-2)=	2.1345e-2(4.52e-2)
Orlraws10p	2.6842e-2(1.56e-2)=	2.8156e-1(6.15e-4)+	7.2753e-1(3.25e-2)+	3.4284e-2(3.44e-2)=	1.7842e-1(4.26e-3)+	4.7015e-1(5.77e-3)+	2.4203e-2(1.54e-2)=	2.2514e-2(4.45e-3)
Brain_Tumor_2	7.4114e-3(3.57e-3)=	8.4411e-1(5.63e-2)+	3.5561e-2(3.15e-2)+	5.6007e-3(5.12e-3)=	8.6548e-2(1.03e-3)+	6.9812e-2(5.45e-2)+	2.5184e-2(4.39e-1)+	7.6175e-3(5.09e-3)
CLL_SUB_111	7.3932e-3(7.42e-3)=	8.4067e-1(4.90e-4)+	5.1374e-2(1.51e-2)+	6.9327e-3(1.93e-2)=	7.1785e-2(1.15e-3)+	6.0227e-2(1.23e-1)+	6.2263e-2(4.33e-2)+	3.8836e-3(1.90e-2)
Smk_can_187	1.6724e-2(7.39e-3)=	1.4163e-1(1.20e-2)+	3.2816e-2(7.09e-2)+	3.7121e-2(2.81e-2)+	1.0673e-1(1.67e-4)+	1.0132e-1(5.76e-3)+	1.8839e-2(3.43e-5)=	1.4300e-2(7.45e-3)
Breast_Cancer	1.7312e-2(4.03e-1)+	8.0773e-2(1.12e-5)+	3.1659e-2(3.05e-2)+	1.9911e-2(3.28e-1)+	5.1512e-2(6.14e-2)+	1.8452e-2(3.10e-1)+	5.5836e-2(9.17e-2)+	5.3162e-3(4.12e-3)
+/-/=	8/0/8	16/0/0	10/0/6	4/2/10	10/1/5	14/0/2	6/0/10	-
Rank2	3.7735	7.4256	4.4250	2.3375	4.3000	5.5475	3.4016	2.1645

datasets, DR-RPMODE outperforms other algorithms. For example, in the TOX_171 dataset, DR-RPMODE quickly reduces Macro F1 Loss by adding fewer features, demonstrating that significant performance improvements occur as the feature subset grows. This indicates that DR-RPMODE has a significant advantage in balancing feature selection and classification performance.

5.2. Analysis of IGD and HV values

Tables 4 and 5 show the average HV and IGD values calculated on the test sets after 30 independent runs of the eight compared algorithms across 16 datasets, with the best average values on each test set marked. Higher HV values indicate better performance, while lower IGD values indicate better performance. The symbols ‘+’ or ‘-’ indicate whether the results obtained by DR-RPMODE are significantly better or worse, respectively than the compared algorithms according to the Wilcoxon significance test. ‘=’ indicates no significant difference between the two algorithms. In other words, the more ‘+’ symbols attached to other algorithms, the better the performance of the proposed algorithm.

In terms of HV, as shown in Table 4, DR-RPMODE achieves the optimal value on 9 out of 16 test sets, while DAEA achieves the optimal value on 4 test sets, NSGAII, MOEA-NZD and MFFS each achieve optimal values on 2 test sets, and RPMODE achieves optimal values on 1 test dataset. Although DR-RPMODE does not reach the optimal value on 7 test sets, the difference with the best-performing algorithms is relatively small, with no significant differences. Additionally, The Friedman test results in Table 4 show that DR-RPMODE ranks first among all comparison algorithms, indicating its overall superiority in HV performance. In terms of IGD, as shown in Table 5, overall, DR-RPMODE still performs well on most test sets, with the minimum IGD

value achieved on 8 out of 16 test sets. DAEA achieves the optimal values on 5 test sets, MFFS and RPMODE each achieve optimal value on 2 test sets, NSGAII and MOEA-NZD achieve the optimal values on 1 test set. It is evident that across all 16 test sets, DR-RPMODE outperforms MOEA/D on every test set and surpasses NSGA-II, MOEA-NZD, MFFS, and PRDH on most of the test sets. Additionally, DR-RPMODE outperforms DAEA on 4 test sets and RPMODE on 6 test sets. The rank values in the last row of the table also indicate that DR-RPMODE ranks first overall.

DR-RPMODE performs well in both HV and IGD, providing further confirmation of the effectiveness of the DR strategy, preference handling technique, and redundancy handling technique. Notably, while several advanced algorithms achieve similar results on low-dimensional test sets, the advantage of DR-RPMODE becomes more evident as the dimensionality of the test sets increases, especially on the four highest-dimensional test sets. This indicates that DR-RPMODE excels in higher dimensions, demonstrating its effectiveness in handling complex, high-dimensional data. The freezing and activation strategies in DR efficiently eliminate irrelevant features, simplifying the problem and allowing RPMODE to focus on the most promising regions, ensuring faster convergence and preserving solution diversity.

5.3. Macro F1 score

Macro F1 score is a crucial criterion for evaluating the classification performance of feature selection algorithms. Table 6 presents the comparison results of the eight compared algorithms in terms of their optimal Macro F1 scores. Among all 16 test sets, the DR-RPMODE algorithm achieves the highest score on 13 test sets, with a perfect score of 1 on 10 test sets, indicating flawless prediction accuracy. Notably,

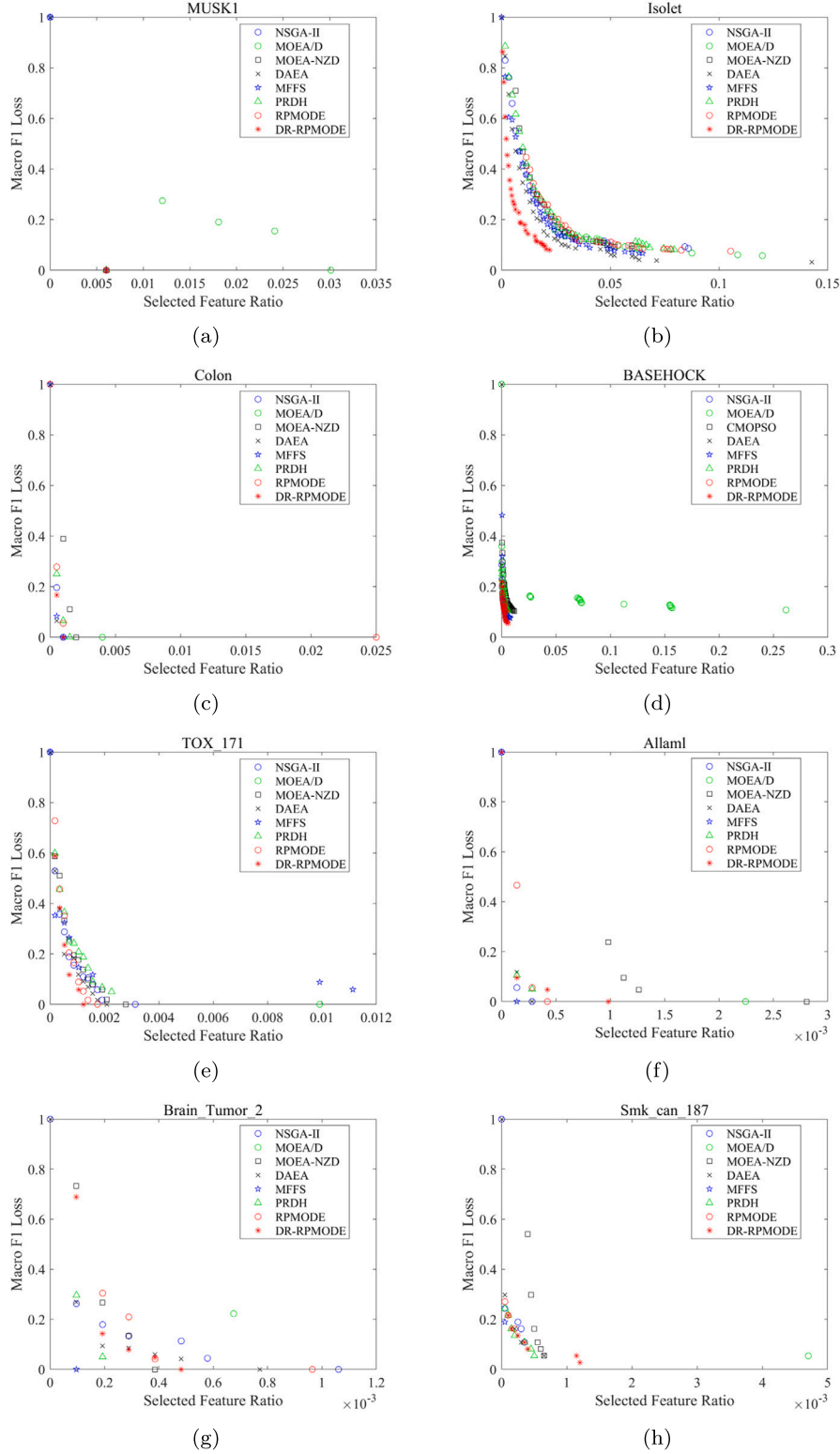


Fig. 7. Distributions of median Pareto front obtained by eight algorithms on the Part of test sets.

DR-RPMODE consistently outperforms other algorithms, particularly on high-dimensional test sets. For example, on the Breast_Cancer dataset, DR-RPMODE achieves a perfect score of 1, whereas other algorithms do not. The average ranking based on the Friedman test indicates that DR-RPMODE ranks first among the algorithms, further reinforcing

its superiority in multi-class classification tasks. The superior performance of DR-RPMODE can be attributed to two key factors: The fast dimensionality reduction strategy DR, which enhances convergence by efficiently eliminating irrelevant features; and the preference handling technique improves classification performance. However, on

Table 6
Comparative results of Macro F1 score obtained by eight algorithms on the test sets.

Problem	NSGAI	MOEA/D	MOEA-NZD	DAEA	MFFS	PRDH	RPMODE	DR-RPMODE
MUSK1	1	1	1	1	1	1	1	1
Madelon	0.8550	0.7185	0.7233	0.8842	0.8800	0.8692	0.8850	0.8875
Isolet	0.9081	0.9209	0.9489	0.9626	0.9573	0.9359	0.8974	0.9562
Orl	1	0.9958	0.9916	1	0.9620	0.9620	0.9831	1
Colon	1	0.9722	1	1	1	1	1	1
Lung	0.9917	0.9667	0.9750	0.9833	0.9917	0.9917	1	1
Basehock	0.8537	0.9012	0.9138	0.8825	0.9228	0.8586	0.9315	0.9464
9tumor	0.8333	0.8056	0.8611	0.9165	0.8333	0.8778	0.8746	0.9044
Tox_171	0.9902	0.9775	1	1	0.9658	0.9852	1	1
Drivface	0.9253	0.9525	0.9242	1	0.9333	0.9447	0.9502	1
Allaml	1	0.9524	1	1	1	1	1	1
Orlraws10p	1	1	1	1	0.9772	1	1	1
Brain_Tumor_2	1	0.9667	1	1	1	1	0.9556	1
CLL_SUB_111	0.9597	0.9717	0.8811	0.9371	0.8938	0.9001	0.9242	0.9364
Smk_can_187	0.9033	0.8825	0.9279	0.9099	0.8180	0.9359	0.8182	0.9455
Breast_Cancer	0.9545	0.9025	0.9256	0.9848	0.8781	0.9242	0.9568	1
Rank3	4.1733	4.9542	3.7394	2.5188	3.8836	3.5127	2.8756	1.8115

Table 7
Comparative results of AFN values obtained by eight algorithms on the test sets.

Problem	NSGAI	MOEA/D	MOEA-NZD	DAEA	MFFS	PRDH	RPMODE	DR-RPMODE
MUSK1	1.0000	1.6767	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Madelon	7.3333	25.333	11.6667	8.6667	8.3333	7.0000	9.6667	6.0000
Isolet	51.0000	54.3333	61.6667	52.3333	64.3333	65.3333	54.3333	33.1500
Orl	29.3333	44.3333	30.6667	28.3333	28.6667	18.3333	110.6667	51.3333
Colon	2.3333	25.6667	5.3333	3.0000	2.3333	3.0000	22.3333	2.3333
Lung	10.3333	98.3333	11.0000	13.3333	5.3333	10.3333	9.0000	15.0000
Basehock	17.6667	144.0000	32.4682	31.0000	46.6667	15.6667	25.6667	26.8446
9tumor	11.0000	110.0000	12.0000	26.0000	35.6667	5.3333	20.6667	6.6667
Tox_171	21.6667	130.0000	34.0000	11.3333	31.0000	17.0000	14.3333	8.7333
Drivface	10.6667	63.6667	7.0000	7.3333	7.6667	7.3333	16.6667	16.3902
Allaml	2.0000	60.3333	4.3333	2.3333	1.0000	2.0000	2.6667	4.3333
Orlraws10p	4.3333	36.3333	8.3333	3.0000	21.3333	3.0000	2.6667	2.3333
Brain_Tumor_2	12.2140	100.2832	7.7141	14.1425	1.9285	7.0713	7.7141	4.4998
CLL_SUB_111	15.2798	183.3573	24.6827	12.9290	1.7631	5.8768	27.6211	10.5783
Smk_can_187	12.3333	152.3333	17.6667	10.3333	1.0000	5.3333	19.3333	40.0000
Breast_Cancer	5.7167	130.6678	7.3501	11.1612	8.5338	7.3501	15.5168	11.9779
Rank4	3.3882	7.4021	3.7244	4.0133	3.5295	3.2392	4.7103	3.6493

some datasets such as Isolet, DR-RPMODE does not achieve the optimal score. This is likely due to FR and AN parameters settings, which may not have been well-tuned for these datasets.

5.4. The number of selected features

Another important criterion for evaluating the performance of an algorithm is the number of features in the subset that achieves the minimum Macro F1 Loss. Table 7 presents the comparison results of the eight compared algorithms in terms of AFN. Among all 16 test sets, the DR-RPMODE algorithm achieves the minimum AFN on 6 test sets. According to the Friedman test's average rankings, PRDH ranks first among all algorithms, and although DR-RPMODE ranks fourth, the difference in their ranking values is small. The average AFN of DR-RPMODE is calculated to be 14.14, indicating that DR-RPMODE can maintain a high Macro F1 score while still selecting a relatively small number of features.

The reason why the number of features selected by DR-RPMODE lags behind leading algorithms may be due to its preference handling method that focuses more on classification performance. During environmental selection, solutions with better classification performance are more likely to be retained, while those with poorer classification performance are more likely to be discarded. However, classification performance and the number of selected features are often conflicting objectives. Solutions with better classification performance usually contain more features, while solutions with poorer classification performance contain fewer features. Therefore, it is difficult for DR-RPMODE to select the least number of features while prioritizing classification performance.

5.5. Running time

Table 8 summarizes the rank values of all comparison algorithms across the four aforementioned metrics, with the last row providing the average rank value, representing their overall performance. It can be seen that DR-RPMODE and DAEA rank first and second, with average rank values of 2.6847 and 2.9075, respectively, indicating a small difference between them and a significant lead over other algorithms. Therefore, we only compare the runtime of DR-RPMODE and DAEA. Table 9 presents the average runtime of DR-RPMODE and DAEA on 16 datasets. DAEA has a shorter average runtime on datasets with lower feature dimensions. This is expected because DAEA has a simpler search space in lower dimensions, requiring fewer computational resources and evaluations to reach a near-optimal solution. In contrast, DR-RPMODE involves more sophisticated dimensionality reduction steps, which might take longer on datasets with fewer features but are still efficient overall.

However, DR-RPMODE has a shorter runtime on datasets with higher dimensions. This demonstrates the efficiency of DR-RPMODE in handling high-dimensional data. The key to DR-RPMODE's performance in high-dimensional cases is its fast dimensionality reduction strategy DR. By efficiently reducing the feature space at the outset, DR-RPMODE simplifies the optimization task, allowing the algorithm to concentrate on more promising regions of the solution space. This approach leads to faster convergence and reduced runtime, especially on high-dimensional datasets, without compromising solution quality.

Table 8

Summary of rank values of comparison algorithms on HV, IGD, Macro F1 score, AFN.

	NSGAI	MOEA/D	MOEA-NZD	DAEA	MFFS	PRDH	RPMODE	DR-RPMODE
Rank1	3.4071	7.5516	3.8438	2.7507	4.1128	4.4840	4.2754	2.1135
Rank2	3.7735	7.4256	4.4250	2.3375	4.3000	5.5475	3.4016	2.1645
Rank3	4.1733	4.9542	3.7394	2.5188	3.8836	3.5127	2.8756	1.8115
Rank4	3.3882	7.4021	3.7244	4.0133	3.5295	3.2392	4.7103	3.6493
Average	3.6855	6.0834	3.9332	2.9075	4.0065	4.4458	3.5657	2.6847

Table 9

Comparative results of Running time values obtained by DAEA, DR-RPMODE.

Problem	DAEA	DR-RPMODE
MUSK1	6.0554e+1 (1.15e+1)	9.4215e+1 (1.22e+1)
Madelon	1.7753e+2 (2.54e+1)	2.1825e+2 (3.01e+1)
Isolet	1.9853e+2 (3.54e+1)	2.8743e+2 (2.71e+1)
Orl	7.7205e+1 (1.75e+1)	1.3848e+2 (1.66e+1)
Colon	6.9451e+1 (1.64e+1)	8.9425e+1 (1.74e+1)
Lung	7.6524e+1 (1.82e+1)	1.4823e+2 (1.67e+1)
Basehock	3.1224e+2 (4.10e+1)	4.8243e+2 (7.34e+1)
9tumor	6.5405e+1 (1.24e+1)	7.8634e+1 (1.73e+1)
Tox_171	8.8538e+1 (1.89e+1)	1.0761e+2 (7.83e+0)
Drivface	1.1524e+2 (2.57e+1)	1.0475e+2 (2.66e+1)
Allaml	9.8422e+1 (1.02e+1)	8.8524e+1 (6.54e+0)
Orlraws10p	1.2445e+2 (1.72e+1)	1.1153e+2 (8.70e+0)
Brain_Tumor_2	1.2544e+2 (2.64e+1)	1.0576e+2 (8.35e+0)
CLL_SUB_111	1.2842e+2 (5.30e+0)	9.6507e+1 (8.84e+0)
Smk_can_187	1.9857e+2 (3.44e+1)	1.2754e+2 (2.25e+1)
Breast_Cancer	2.156e+2 (2.57e+0)	1.3228e+2 (1.43e+1)

6. Further analysis

6.1. Sensitivity analysis of FR_{max}

To verify the effectiveness of setting $FR_{max} = 0.3$, we compare its performance against other values of FR_{max} in terms of the HV. Table 10 summarizes the average HV results obtained by the proposed algorithm for various datasets with different settings of FR_{max} . As shown in Table 10, the setting $FR_{max} = 0.3$ consistently yields the highest HV scores across most datasets, demonstrating its superior ability to achieve better Pareto front approximations compared to other tested values of FR_{max} . This analysis supports the choice of $FR_{max} = 0.3$ as an effective parameter for balancing exploration and exploitation in our algorithm.

6.2. Effectiveness of the key components

The redundancy handling and preference handling techniques are the key components of DR-RPMODE. To verify the effectiveness of the redundancy handling and preference handling techniques, we separately compare the performance of DR-RPMODE with and without the redundancy handling mechanism (denoted as DR-PMODE) and with and without the preference handling mechanism (denoted as DR-RMODE). The average HV results obtained by DR-PMODE, DR-RMODE, and DR-RPMODE on the 16 datasets over 30 runs are listed in Table 11. As shown in the table, DR-RPMODE consistently outperforms the other approaches across all datasets, demonstrating the effectiveness of the proposed redundancy and preference handling mechanisms.

6.3. Computational complexity

The computational complexity of MOEAs such as DAEA is given by $O(it_{max}(D \times NP + C \times NP))$, where it_{max} represents the maximum number of iterations, D is the number of dataset dimensions (i.e., number of features), C is the computational cost of the evaluation function, and NP is the population size. The computational complexity of the DR-RPMODE algorithm is given by $O(DR_{it}(D + C) + RPMODE_{it}(D \times NP + C \times NP))$, where DR_{it} is the number of iterations performed by the

DR algorithm, and $RPMODE_{it}$ is the number of iterations performed by the RPMODE algorithm. Additionally, it is important to note that $it_{max} \times NP = DR_{it} + RPMODE_{it} \times NP$.

Since DR implements fast dimensionality reduction and RPMODE performs the search in a smaller space, the values of D and C in RPMODE are smaller. Additionally, as the dimensionality of the dataset increases, the DR algorithm removes more irrelevant features, which further reduces the evaluation cost. In other words, the values of D and C decrease more significantly, making the computational cost of the DR-RPMODE algorithm increasingly advantageous compared to other MOEAs. This is demonstrated in Table 9, which shows the running times of the algorithms across different datasets.

7. Conclusion

In this paper, we propose DR-RPMODE, a hybrid algorithm for feature selection in high-dimensional data. DR reduces dimensionality by eliminating irrelevant features, while RPMODE addresses redundancy and prioritizes classification performance. Experimental results show that DR-RPMODE outperforms other algorithms in HV, IGD, and classification performance, with its advantages becoming more apparent as data dimensionality increases, demonstrating its scalability. The algorithm has potential applications in bioinformatics, image processing, and text mining.

However, despite its efficiency, the proposed algorithm has some limitations that need to be addressed in future research. Firstly, DR-RPMODE achieves the shortest runtime only on ultra-high-dimensional datasets, while its runtime is less competitive on lower-dimensional datasets. Secondly, its performance is sensitive to parameters such as FR and AN . Suboptimal parameter settings may lead to the removal of relevant features, as observed on datasets like Isolet. In future work, we will focus on improving runtime on low-dimensional datasets while maintaining classification performance. Additionally, we will conduct in-depth experiments to analyze parameter sensitivity and optimize settings for various datasets. Further studies will also evaluate DR-RPMODE on more high-dimensional categorical datasets and extend it to regression problems, aiming to enhance its performance, generality, and stability across real-world applications.

CRedit authorship contribution statement

Xuezhi Yue: Writing – review & editing, Project administration, Methodology, Funding acquisition. **Yihang Liao:** Writing – original draft, Validation, Methodology, Conceptualization. **Hu Peng:** Writing – review & editing, Methodology, Funding acquisition. **Lanlan Kang:** Validation, Funding acquisition. **Yuan Zeng:** Validation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Table 10

Comparative results of HV values (mean and standard deviation) obtained by $FR_{max} = 0.2$, $FR_{max} = 0.3$, and $FR_{max} = 0.4$.

Problem	$FR_{max} = 0.2$	$FR_{max} = 0.4$	$FR_{max} = 0.3$
MUSK1	1.7355e-1(0.00e+0)=	1.5781e-1(2.34e-2)+	1.7355e-1(0.00e+0)
Madelon	6.7244e-1 (2.13e-2)=	6.6238e-1 (3.08e-2)=	6.7257e-1 (3.45e-2)
Isolet	8.5782e-1(3.24e-2)=	8.4149e-1(1.72e-2)=	8.5548e-1(2.57e-2)
Orl	7.2744e-1(1.32e-2)=	7.3015e-1(2.35e-2)=	7.3128e-1(2.32e-2)
Colon	5.7217e-1(1.38e-2)=	5.7226e-1(1.47e-2)=	5.7305e-1(1.73e-2)
Lung	7.3754e-1(1.92e-2)=	7.3436e-1(2.28e-2)=	7.4377e-1(1.36e-2)
Basehock	9.4755e-1 (1.50e-1)=	9.4736e-1 (1.62e-1)=	9.4933e-1 (1.26e-1)
9tumor	6.1172e-1(1.48e-2)=	6.1305e-1(2.31e-2)=	6.2177e-1(1.42e-2)
Tox_171	6.7255e-1(2.13e-2)=	6.7560e-1(2.41e-2)=	6.7490e-1(2.72e-2)
Drivface	7.8832e-1(2.58e-1)=	7.9014e-1(2.04e-1)=	7.9563e-1(2.68e-1)
Allaml	7.8326e-1(4.02e-2)=	7.8804e-1(2.76e-2)=	7.9165e-1(3.14e-2)
Orlraws10p	3.9544e-1(5.31e-3)+	4.1473e-1(7.52e-3)=	4.2352e-1(9.32e-3)
Brain_Tumor_2	8.7540e-1(7.43e-3)=	8.6882e-1(6.91e-3)=	8.7782e-1(6.36e-3)
CLL_SUB_111	7.7565e-1(1.57e-2)+	7.8133e-1(2.46e-2)+	7.8353e-1(2.48e-2)
Smk_can_187	8.3701e-1(7.11e-2)+	8.4243e-1(8.28e-2)=	8.5013e-1(7.17e-2)
Breast_Cancer	8.8856e-1(2.77e-2)+	9.1718e-1(1.84e-2)=	9.2218e-1(1.73e-2)

Table 11

Comparative results of HV values (mean and standard deviation) obtained by DR-PMODE, DR-RMODE, and DR-PRMODE on the test sets.

Problem	DR-PMODE	DR-RMODE	DR-PRMODE
MUSK1	1.7355e-1(0.00e+0)=	1.7355e-1(0.00e+0)=	1.7355e-1(0.00e+0)
Madelon	6.3752e-1 (1.42e-2)=	6.4208e-1 (2.43e-2)=	6.7257e-1 (3.45e-2)
Isolet	7.8042e-1 (7.46e-3)+	7.8513e-1 (2.24e-3)+	8.5548e-1(2.57e-2)
Orl	7.3014e-1 (9.51e-3)=	7.2644e-1 (1.48e-2)=	7.3128e-1(2.32e-2)
Colon	5.6217e-1 (8.42e-3)=	5.7024e-1 (2.57e-2)=	5.7305e-1(1.73e-2)
Lung	7.1145e-2 (4.24e-2)=	7.2564e-1 (8.96e-3)=	7.4377e-1(1.36e-2)
Basehock	9.4047e-1 (1.64e-1)=	9.0428e-1 (1.33e-1)+	9.4933e-1 (1.26e-1)
9tumor	5.3544e-1 (5.33e-2)+	6.1105e-1 (4.27e-2)=	6.2177e-1(1.42e-2)
Tox_171	6.2438e-1 (3.42e-3)+	6.5542e-1 (5.04e-2)+	6.7490e-1(2.72e-2)
Drivface	7.5705e-1(2.33e-1)+	7.5184e-1(1.49e-1)+	7.9563e-1(2.68e-1)
Allaml	7.9151e-1(4.14e-2)=	7.9124e-1(3.55e-2)=	7.9165e-1(3.14e-2)
Orlraws10p	4.0483e-1(4.72e-3)=	3.8649e-1(5.22e-3)=	4.2352e-1(9.32e-3)
Brain_Tumor_2	8.3038e-1 (8.40e-3)+	8.2256e-1 (1.54e-2)+	8.7782e-1(6.36e-3)
CLL_SUB_111	7.8186e-1 (2.38e-2)=	7.5722e-1 (3.07e-2)+	7.8353e-1(2.48e-2)
Smk_can_187	8.3441e-1(3.43e-2)=	8.2825e-1(5.42e-2)=	8.5013e-1(7.17e-2)
Breast_Cancer	8.9072e-1 (1.83e-2)=	9.0155e-1 (7.54e-3)=	9.2218e-1(1.73e-2)

Data availability

Data will be made available on request.

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